A03 / INDIVIDUAL LAB REPORT

Conditioning, Sampling & Evaluation

Neural Network Zoo • Neural Network Safari Reflection

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Role Conditioning, Sampling & Evaluation Lead

COMPUTATIONAL OBSERVATION | GENERATIVE MODEL EVALUATION

1. Introduction & Objectives

My primary responsibility for the Neural Network Safari project focused on class-conditioning mechanisms, reverse diffusion sampling loops, and quantitative evaluation metrics.

In the context of our safari metaphor, this phase represents the *observation and wildlife tracking* layer, validating how well our models generate targeted, high-fidelity samples from learned representations.

> **Primary objective**
> Develop and evaluate the conditioning and sampling components that guide generation toward target classes while providing quantitative and visual methods for examining model outputs.

2. Technical Execution & Methodology

To complete our model evaluation and generation pipeline, I implemented the following components.

01 Class-Conditioning Mechanisms

Engineered embedding lookup tables and injection layers to guide generation toward specific target classes, including MNIST digit categories.

02 Reverse Sampling Loops

Coded iterative denoising loops that transition from pure Gaussian noise x_T back to clean generated images x_0.

Sampling pathway

$$x_T \longrightarrow x_{T-1} \longrightarrow \cdots \longrightarrow x_1 \longrightarrow x_0$$

The reverse process progressively transforms noisy latent samples into generated outputs.

03 Visualization Grids

Built Matplotlib-based plotting scripts to output multi-class visual sampling grids at designated training epochs.

04 Quantitative Metrics

Integrated evaluation functions to monitor loss convergence and output variance across generation batches.

3. Challenges & Solutions

Challenge: Sampling instability
High variance and noisy outputs during early-stage reverse sampling iterations made it difficult to obtain consistently clean generated samples.

Solution: Noise-step tuning and clipping
Tuned the noise-schedule step-size parameters and implemented clipping routines to stabilize sample trajectories during the final denoising steps.

4. Reflection

Working on sampling and conditioning demonstrated how crucial inference-time controls are for generative models. Conditioning mechanisms influence which classes the model is guided to generate, while reverse sampling determines how noisy states are progressively transformed into final outputs.

Collaborating closely with Olu on the U-Net backbone allowed us to seamlessly integrate class embeddings and produce clean, reliable classification and generation results.

Key insight
A generative model must be evaluated not only by what it learns, but by what it produces during inference.
Conditioning, reverse sampling, and quantitative evaluation work together to assess whether generated outputs align with the intended target and remain consistent across sampling batches.

TAKI | A03 NEURAL NETWORK ZOO

Conditioning • Reverse Sampling • Quantitative Evaluation